

Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an adherent point of A if every open neighbourhood U of x satisfies

$$U \cap A \neq \emptyset.$$

In the special case where X is a metric space with metric d , this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \emptyset.$$

Definition 2. Let X be a topological space and $A \subseteq X$. The set A is called dense in X if every nonempty open set $U \subseteq X$ satisfies

$$U \cap A \neq \emptyset.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. Let X be a topological space and $A \subseteq X$. The set A is called nowhere dense in X if

$$\text{int}(\overline{A}) = \emptyset.$$

Definition 4. A topological space X is separable if: $\exists D \subseteq X : D$ countable and dense.

Baire Category Theorem

Theorem 1 (Baire Category Theorem). Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^{\infty}$ be a sequence of dense open subsets of X . Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X .

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \quad r_1 < 1.$$

Inductively, having chosen y_n, r_n , use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \quad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \rightarrow 0$. Thus (y_n) is Cauchy, so by completeness $y_n \rightarrow y \in X$.

For each fixed n , all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \emptyset,$$

so the intersection is dense. □

Remark 1. From the definitions it follows that a set $B \subseteq X$ satisfies $\text{int}(B) = \emptyset \iff X \setminus B$ is dense in X . In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X .

Corollary 1. Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X . Then

$$\text{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \emptyset.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. □

Finite Products of Metric Spaces

Definition 5. Let (X_k, d_k) , $k = 1, \dots, n$, be metric spaces and set

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric if convergence in (X, d) is coordinatewise,

$$x^{(m)} = (x_1^{(m)}, \dots, x_n^{(m)}) \rightarrow x = (x_1, \dots, x_n)$$

in X if and only if for each $k = 1, \dots, n$,

$$x_k^{(m)} \rightarrow x_k \quad \text{in } X_k.$$

Theorem 2. Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k .

Proof. First, if $E_k \subset X_k$ is closed for each k , then

$$E_1 \times \cdots \times E_n$$

is closed in X . Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \rightarrow x$ in X , then by coordinatewise convergence, $x_k^{(m)} \rightarrow x_k$ in X_k . Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X , since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^n V^{(k)}$$

is open in X , and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \dots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \quad k = 1, \dots, n.$$

Then $x_k^{(m)} \rightarrow x_k$ for each k , hence $x^{(m)} \rightarrow x$ in X . Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. $\square$

Theorem 3 (Heine–Borel Theorem). *For a subspace $E \subseteq \mathbb{R}^n$, the following are equivalent:*

- (i) E is compact;
- (ii) every sequence in E has a convergent subsequence with limit in E ;
- (iii) E is closed and bounded.

Proof. By the general compactness theorem for metric spaces, for any metric space the following are equivalent: compactness, sequential compactness, and completeness together with total boundedness.

Now let $E \subseteq \mathbb{R}^n$.

If E is closed and bounded, then E is complete (since a subspace of $\mathbb{R}^n$ is complete if and only if it is closed) and totally bounded (since a subset of $\mathbb{R}^n$ is totally bounded if and only if it is bounded). Hence E is compact, and therefore every sequence in E has a convergent subsequence.

Conversely, if E is compact, then by the same general theorem E is complete and totally bounded. Since E is a subspace of $\mathbb{R}^n$, completeness implies that E is closed, and total boundedness implies that E is bounded. Thus E is closed and bounded.

Therefore (i), (ii), and (iii) are equivalent. $\square$

Theorem 4 (Uniform Continuity Theorem). *Let (X, d) and (Y, ρ) be metric spaces, and suppose that X is compact. Then every continuous function $f : X \rightarrow Y$ is uniformly continuous.*

Proof. Suppose f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for every $m \in \mathbb{N}$ there are points $x_m, z_m \in X$ with

$$d(x_m, z_m) < \frac{1}{m} \quad \text{but} \quad \rho(f(x_m), f(z_m)) \geq \varepsilon.$$

Since X is compact, the sequence (x_m) has a convergent subsequence; passing to that subsequence, we may assume

$$x_m \rightarrow x \in X.$$

Because $d(x_m, z_m) \rightarrow 0$, also

$$z_m \rightarrow x.$$

By continuity of f at x ,

$$f(x_m) \rightarrow f(x) \quad \text{and} \quad f(z_m) \rightarrow f(x).$$

Hence

$$\rho(f(x_m), f(z_m)) \leq \rho(f(x_m), f(x)) + \rho(f(x), f(z_m)) \rightarrow 0,$$

which contradicts

$$\rho(f(x_m), f(z_m)) \geq \varepsilon.$$

Therefore f is uniformly continuous. □

Theorem 5 (Continuity and boundedness of linear operators). *Let X and Y be normed spaces and let $T : X \rightarrow Y$ be linear. The following are equivalent:*

- (i) T is continuous;
- (ii) T is continuous at 0;
- (iii) T is bounded, i.e. there exists $C > 0$ such that

$$\|T(x)\| \leq C\|x\| \quad \text{for all } x \in X.$$

Proof. (i) $\Rightarrow$ (ii) is immediate.

(ii) $\Rightarrow$ (iii): Since T is continuous at 0, there exists $\delta > 0$ such that

$$\|x\| \leq \delta \Rightarrow \|T(x)\| \leq 1.$$

For arbitrary $x \neq 0$, set $y = \frac{\delta}{\|x\|}x$, so $\|y\| = \delta$ and hence

$$\|T(y)\| \leq 1.$$

By linearity,

$$\|T(x)\| = \frac{\|x\|}{\delta} \|T(y)\| \leq \frac{1}{\delta} \|x\|.$$

Thus T is bounded.

(iii) $\Rightarrow$ (i): If $\|T(x)\| \leq C\|x\|$, then for all $x, x_0 \in X$,

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\| \leq C\|x - x_0\|,$$

so T is Lipschitz, hence continuous. □

Theorem 6 (Principle of Uniform Boundedness). *Let X and Y be Banach spaces and let $\{T_\alpha\}_{\alpha \in A}$ be a family of continuous linear operators $T_\alpha : X \rightarrow Y$. Suppose that for each $x \in X$,*

$$\sup_{\alpha \in A} \|T_\alpha x\| < \infty.$$

Then

$$\sup_{\alpha \in A} \|T_\alpha\| < \infty.$$

Proof. For $n \in \mathbb{N}$, define

$$E_n := \left\{ x \in X : \sup_{\alpha \in A} \|T_\alpha x\| \leq n \right\}.$$

Then each E_n is closed, $E_n \subseteq E_{n+1}$, and by assumption

$$X = \bigcup_{n=1}^{\infty} E_n.$$

By the Baire Category Theorem, some E_m has nonempty interior. Thus there exist $x_0 \in X$ and $\varepsilon > 0$ such that

$$B(x_0, \varepsilon) \subseteq E_m.$$

Let $c := \sup_{\alpha \in A} \|T_\alpha x_0\| < \infty$. If $\|x\| < 1$, then $x_0 + \varepsilon x \in E_m$, hence

$$\|T_\alpha(x_0 + \varepsilon x)\| \leq m.$$

Thus

$$\varepsilon \|T_\alpha x\| = \|T_\alpha(x_0 + \varepsilon x) - T_\alpha(x_0)\| \leq m + c.$$

Therefore

$$\|T_\alpha x\| \leq \frac{m+c}{\varepsilon} \quad \text{for all } \|x\| < 1.$$

Hence $\|T_\alpha\| \leq \frac{m+c}{\varepsilon}$ for all $\alpha \in A$, and so

$$\sup_{\alpha \in A} \|T_\alpha\| < \infty.$$

□

Topologies

Definition 6. *Let X be a set. A topology on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that*

(i) $\emptyset \in \mathcal{T}$ and $X \in \mathcal{T}$,

(ii) if $\{U_\alpha\}_{\alpha \in I} \subseteq \mathcal{T}$, then

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{T},$$

(iii) if $U_1, \dots, U_n \in \mathcal{T}$, then

$$\bigcap_{i=1}^n U_i \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is called a topological space. The sets $U \in \mathcal{T}$ are called open sets. A subset $F \subseteq X$ is called closed if its complement is open.

Separation Axioms

Definition 7 (T_0 space). A topological space $(X, \mathcal{T})$ is called a T_0 -space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that

$$x \in U, y \notin U \quad \text{or} \quad y \in U, x \notin U.$$

Definition 8 (T_1 space). A topological space $(X, \mathcal{T})$ is called a T_1 -space if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that

$$y \in U \quad \text{and} \quad x \notin U.$$

Equivalently, $(X, \mathcal{T})$ is T_1 if and only if every singleton set $\{x\}$ is closed in X .

Definition 9 (T_2 space / Hausdorff). A topological space $(X, \mathcal{T})$ is called a T_2 -space, or Hausdorff, if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that

$$x \in U, \quad y \in V, \quad U \cap V = \emptyset.$$

Definition 10 (Regular space). A topological space $(X, \mathcal{T})$ is called regular if for every closed set $F \subseteq X$ and every point $x \in X \setminus F$, there exist open sets $U, V \in \mathcal{T}$ such that

$$x \in U, \quad F \subseteq V, \quad U \cap V = \emptyset.$$

Definition 11 (T_3 space). A topological space $(X, \mathcal{T})$ is called a T_3 -space if it is both T_1 and regular.

Definition 12 (Normal space). A topological space $(X, \mathcal{T})$ is called normal if for any two disjoint closed sets $F, G \subseteq X$, there exist open sets $U, V \in \mathcal{T}$ such that

$$F \subseteq U, \quad G \subseteq V, \quad U \cap V = \emptyset.$$

Definition 13 (T_4 space). A topological space $(X, \mathcal{T})$ is called a T_4 -space if it is both T_1 and normal.

Theorem 7. Every compact Hausdorff space is normal. In particular, every compact Hausdorff space is a T_4 -space.

Proof. Let $F, G \subseteq X$ be disjoint closed sets. Since X is Hausdorff, for each $x \in F$ and $y \in G$ there exist disjoint open sets $U_{x,y}$ and $V_{x,y}$ such that

$$x \in U_{x,y}, \quad y \in V_{x,y}, \quad U_{x,y} \cap V_{x,y} = \emptyset.$$

For fixed $x \in F$, the sets $\{V_{x,y}\}_{y \in G}$ form an open cover of G . Since G is closed in the compact space X , it is compact, so we may choose a finite subcover

$$G \subseteq V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Let

$$U_x := U_{x,y_1} \cap \cdots \cap U_{x,y_n}, \quad V_x := V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Then $x \in U_x$, $G \subseteq V_x$, and $U_x \cap V_x = \emptyset$.

The family $\{U_x\}_{x \in F}$ is an open cover of F . Since F is compact, choose a finite subcover

$$F \subseteq U_{x_1} \cup \cdots \cup U_{x_m}.$$

Define

$$U := U_{x_1} \cup \cdots \cup U_{x_m}, \quad V := V_{x_1} \cap \cdots \cap V_{x_m}.$$

Then U and V are open, $F \subseteq U$, $G \subseteq V$, and $U \cap V = \emptyset$. Hence X is normal. $\square$

Remark 2 (Continuity criterion in metric spaces). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. A function $f : X \rightarrow Y$ is continuous at $x \in X$ if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that*

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Equivalently, $f(U) \subseteq B(f(x), \varepsilon)$.

Theorem 8 (Uniform limit preserves continuity). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. Suppose (f_n) is a sequence of functions $f_n : X \rightarrow Y$ converging uniformly to $f : X \rightarrow Y$. If each f_n is continuous at $x \in X$, then f is continuous at x .*

Proof. We first recall the following characterization of continuity at a point: f is continuous at x if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Now let $\varepsilon > 0$. By uniform convergence, there exists $N \in \mathbb{N}$ such that

$$d(f_N(y), f(y)) < \frac{\varepsilon}{3} \quad \text{for all } y \in X.$$

Since f_N is continuous at x , there exists a neighbourhood U of x such that

$$d(f_N(y), f_N(x)) < \frac{\varepsilon}{3} \quad \text{for all } y \in U.$$

Then for all $y \in U$,

$$d(f(y), f(x)) \leq d(f(y), f_N(y)) + d(f_N(y), f_N(x)) + d(f_N(x), f(x)) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

By the above characterization, f is continuous at x . □

Theorem 9. *Every metric space (X, d) is a T_4 -space.*

Proof. First, for each $x \in X$,

$$\{x\} = \bigcap_{r>0} \overline{B(x, r)},$$

so singletons are closed and X is T_1 .

Let $E, F \subseteq X$ be disjoint closed sets. For each $x \in E$, there exists $r(x) > 0$ such that

$$B(x, r(x)) \cap F = \emptyset.$$

Similarly, for each $y \in F$, there exists $r(y) > 0$ such that

$$B(y, r(y)) \cap E = \emptyset.$$

Define

$$U := \bigcup_{x \in E} B\left(x, \frac{r(x)}{2}\right), \quad V := \bigcup_{y \in F} B\left(y, \frac{r(y)}{2}\right).$$

Then U, V are open, $E \subseteq U$, $F \subseteq V$.

Suppose $z \in U \cap V$. Then there exist $x \in E$, $y \in F$ such that

$$d(x, z) < \frac{r(x)}{2}, \quad d(y, z) < \frac{r(y)}{2}.$$

Hence

$$d(x, y) \leq d(x, z) + d(z, y) < \frac{r(x) + r(y)}{2} \leq \max\{r(x), r(y)\}.$$

Thus either $d(x, y) < r(x)$ or $d(x, y) < r(y)$, contradicting the choice of $r(x)$ or $r(y)$. Therefore $U \cap V = \emptyset$, and X is normal. □

Basis & Subbasis

Definition 14. Let X be a set. A collection $\mathcal{B}$ of subsets of X is called a *basis* for a topology on X if:

- (i) For every $x \in X$ there exists $B \in \mathcal{B}$ such that $x \in B$.
- (ii) If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Theorem 10 (Topology generated by a basis). If $\mathcal{B}$ satisfies the two conditions above, then there exists a unique topology $\mathcal{T}$ on X whose open sets are precisely the unions of elements of $\mathcal{B}$. That is,

$$\mathcal{T} = \left\{ \bigcup_{\alpha \in I} B_\alpha : B_\alpha \in \mathcal{B}, I \text{ arbitrary} \right\}.$$

Definition 15. Let X be a set. A collection $\mathcal{S}$ of subsets of X is called a *subbasis* for a topology on X if the union of all sets in $\mathcal{S}$ equals X ,

$$X = \bigcup_{S \in \mathcal{S}} S.$$

The topology generated by $\mathcal{S}$ is obtained by first forming the basis consisting of all finite intersections of elements of $\mathcal{S}$,

$$\mathcal{B} = \{S_1 \cap \cdots \cap S_n : S_i \in \mathcal{S}, n \in \mathbb{N}\},$$

and then taking arbitrary unions of these sets.

The topology generated by the subbasis $\mathcal{S}$ consists of all arbitrary unions of elements of this basis. That is, a set $U \subseteq X$ is open if and only if it can be written as

$$U = \bigcup_{\alpha \in I} \left(\bigcap_{i=1}^{n_\alpha} S_{\alpha,i} \right), \quad S_{\alpha,i} \in \mathcal{S}, \quad n_\alpha \in \mathbb{N}.$$

Theorem 11 (Basis criterion). A collection $\mathcal{B}$ of subsets of X is a basis for a topology on X if and only if:

- (i) For every $x \in X$, there exists $B \in \mathcal{B}$ such that $x \in B$;
- (ii) If $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Proof. ($\Rightarrow$) If $\mathcal{B}$ is a basis, then (i) holds since X is open, and (ii) holds since $B_1 \cap B_2$ is open and can be written as a union of basis elements.

($\Leftarrow$) Suppose (i) and (ii) hold. Let $\mathcal{T}$ be the collection of all unions of sets in $\mathcal{B}$. Then $\emptyset, X \in \mathcal{T}$ and arbitrary unions are in $\mathcal{T}$.

To show closure under finite intersections, let $U, V \in \mathcal{T}$ and $x \in U \cap V$. Then there exist $B_1, B_2 \in \mathcal{B}$ such that

$$x \in B_1 \subseteq U, \quad x \in B_2 \subseteq V.$$

By (ii), there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2 \subseteq U \cap V.$$

Hence

$$U \cap V = \bigcup_{x \in U \cap V} B_3(x),$$

so $U \cap V \in \mathcal{T}$. Thus $\mathcal{T}$ is a topology and $\mathcal{B}$ is a basis. □

Theorem 12 (Lindelöf's Theorem). *Let X be a second-countable topological space. Then every open cover of X admits a countable subcover.*

Proof. Let $\mathcal{B} = \{B_n\}_{n \in \mathbb{N}}$ be a countable basis for X , and let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be an open cover of X .

For each $n \in \mathbb{N}$, since $B_n \subseteq X = \bigcup_{\alpha \in A} U_\alpha$, we may choose $\alpha(n) \in A$ such that

$$B_n \subseteq U_{\alpha(n)}.$$

Then the collection $\{U_{\alpha(n)}\}_{n \in \mathbb{N}}$ is countable.

To see that it covers X , let $x \in X$. Since $\mathcal{B}$ is a basis, there exists n such that $x \in B_n$. By construction, $B_n \subseteq U_{\alpha(n)}$, so $x \in U_{\alpha(n)}$. Hence

$$X = \bigcup_{n=1}^{\infty} U_{\alpha(n)}.$$

Thus $\{U_{\alpha(n)}\}_{n \in \mathbb{N}}$ is a countable subcover. □